

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

Перенаправление ввода-вывода

```
ramielsarhanov@ramielsarhanov:~$ ls /etc/ > file.txt
ramielsarhanov@ramielsarhanov:~$ ls >> file.txt
ramielsarhanov@ramielsarhanov:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
chrony.conf
cifs-utils
```

Рисунок 1: Запись в файл

```
ramielsarhanov@ramielsarhanov:~$  
ramielsarhanov@ramielsarhanov:~$ grep .conf file.txt > conf.txt  
ramielsarhanov@ramielsarhanov:~$ cat conf.txt  
anthy-unicode.conf  
asound.conf  
brltty.conf  
chrony.conf  
cpupower-service.conf  
dconf  
dnsmasq.conf  
dracut.conf  
dracut.conf.d  
fprintd.conf  
fuse.conf  
host.conf  
idmapd.conf  
kdump.conf  
krb5.conf  
krb5.conf.d  
ld.so.conf  
ld.so.conf.d  
libaudit.conf  
libuser.conf  
locale.conf  
logrotate.conf
```

Рисунок 2: Поиск расширения .conf


```
/home/ramielsarhanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage0
3/report/bib/cite.bib
/home/ramielsarhanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage0
4/report/_resources/csl
/home/ramielsarhanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage0
4/report/bib/cite.bib
/home/ramielsarhanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage0
5/report/_resources/csl
/home/ramielsarhanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage0
5/report/bib/cite.bib
/home/ramielsarhanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage0
6/report/_resources/csl
/home/ramielsarhanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage0
6/report/bib/cite.bib
/home/ramielsarhanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/_re
sources/csl
/home/ramielsarhanov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/bib
/cite.bib
/home/ramielsarhanov/git-extended/.git/hooks/commit-msg.sample
/home/ramielsarhanov/git-extended/.git/objects/29/cc092cb27b095c44b1ea04375889cba2e9c173
/home/ramielsarhanov/git-extended/.git/objects/cf
/home/ramielsarhanov/git-extended/.git/objects/c8
/home/ramielsarhanov/git-extended/.git/objects/27/cb033de02120b20f9a7477446c734e833f7c25
/home/ramielsarhanov/git-extended/.git/config
/home/ramielsarhanov/conf.txt
ramielsarhanov@ramielsarhanov:~$
```

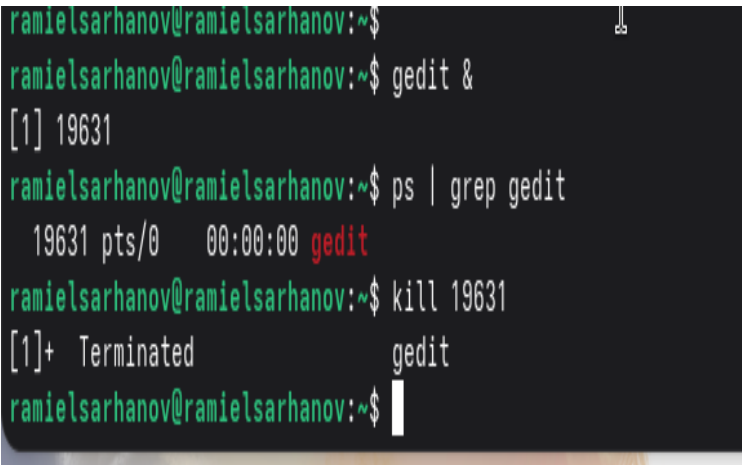
Рисунок 3: Поиск файлов

```
/etc/foomatic/hashe.s.d/hashe.s.new
find: '/etc/libvirt': Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: /etc/logrotate.d/httpd
'/etc/lvm/archive': Permission denied
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sss': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
/etc/sysconfig/htcacheclean
/etc/systemd/system/httpd.service.d
:
```

Рисунок 4: Поиск файлов

```
ramielsarhanov@ramielsarhanov:~$  
ramielsarhanov@ramielsarhanov:~$ find /etc -name "h*" -print | less  
ramielsarhanov@ramielsarhanov:~$  
ramielsarhanov@ramielsarhanov:~$  
ramielsarhanov@ramielsarhanov:~$ find ~ -name "log*" > logfile &  
[1] 19547  
ramielsarhanov@ramielsarhanov:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
ramielsarhanov@ramielsarhanov:~$ rm logfile  
ramielsarhanov@ramielsarhanov:~$  
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 5: Фоновый запуск процесса



```
ramielsarhanov@ramielsarhanov:~$  
ramielsarhanov@ramielsarhanov:~$ gedit &  
[1] 19631  
ramielsarhanov@ramielsarhanov:~$ ps | grep gedit  
19631 pts/0    00:00:00 gedit  
ramielsarhanov@ramielsarhanov:~$ kill 19631  
[1]+  Terminated                  gedit  
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
df(1)                                User Commands                                df(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount of space available on the
    file system containing each file name argument.  If no file name is given, the space available on all
    currently mounted file systems is shown.  Space is shown in 1K blocks by default, unless the environ-
    ment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a mounted file system, df shows
    the space available on that file system rather than on the file system containing the device node.
    This version of df cannot show the space available on unmounted file systems, because on most kinds of
    systems doing so requires non-portable intimate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all file systems by default.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        include pseudo, duplicate, inaccessible file systems

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes;
        see SIZE format below

    --direct
        show statistics for a file instead of mount point

    -h, --human-readable
        Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
du(1)                                User Commands                                du(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is usually smaller,
        it may be larger due to holes in ('sparse') files, internal fragmentation, indirect blocks, and
        the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes;
        see SIZE format below

    -b, --bytes
        equivalent to '--apparent-size --block-size=1'

    -c, --total
        produce a grand total

Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
ramielsarhanov@ramielsarhanov:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 39575280 108592192  27% /
devtmpfs        4011284      0    4011284   0% /dev
tmpfs           4035656      92    4035564   1% /dev/shm
tmpfs           1614264     1956    1612308   1% /run
tmpfs            1024        0      1024     0% /run/credentials/systemd-journald.service
tmpfs           4035660    136780    3898880   4% /tmp
/dev/nvme0n1p3 155186176 39575280 108592192  27% /home
/dev/nvme0n1p2  1992552    518800    1352512  28% /boot
/dev/loop2       49280     49280        0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop0      115712    115712        0 100% /var/lib/snapd/snap/hugo/25676
/dev/loop1       68480     68480        0 100% /var/lib/snapd/snap/core24/1349
tmpfs            1024        0      1024     0% /run/credentials/systemd-resolved.service
tmpfs           807128      188    806940   1% /run/user/1091
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 9: Справка по команде du

```
4      ./git-extended/.git/objects/26
4      ./git-extended/.git/objects/a2
4      ./git-extended/.git/objects/27
4      ./git-extended/.git/objects/34
76     ./git-extended/.git/objects
8      ./git-extended/.git/logs/refs/heads
8      ./git-extended/.git/logs/refs/remotes/origin
8      ./git-extended/.git/logs/refs/remotes
16     ./git-extended/.git/logs/refs
20     ./git-extended/.git/logs
212    ./git-extended/.git
220    ./git-extended
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
882888 .
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 10: Запуск команды du


```
/home/ramielsarhanov/git-extended/.git/objects/c8
/home/ramielsarhanov/git-extended/.git/objects/53
/home/ramielsarhanov/git-extended/.git/objects/eb
/home/ramielsarhanov/git-extended/.git/objects/77
/home/ramielsarhanov/git-extended/.git/objects/21
/home/ramielsarhanov/git-extended/.git/objects/9c
/home/ramielsarhanov/git-extended/.git/objects/26
/home/ramielsarhanov/git-extended/.git/objects/a2
/home/ramielsarhanov/git-extended/.git/objects/27
/home/ramielsarhanov/git-extended/.git/objects/34
/home/ramielsarhanov/git-extended/.git/logs
/home/ramielsarhanov/git-extended/.git/logs/refs
/home/ramielsarhanov/git-extended/.git/logs/refs/heads
/home/ramielsarhanov/git-extended/.git/logs/refs/remotes
/home/ramielsarhanov/git-extended/.git/logs/refs/remotes/origin
/home/ramielsarhanov/monthly
/home/ramielsarhanov/reports
/home/ramielsarhanov/reports/monthly
/home/ramielsarhanov/reports/monthly/monthly
/home/ramielsarhanov/ski.plases
/home/ramielsarhanov/ski.plases/equipment
/home/ramielsarhanov/ski.plases/plans
/home/ramielsarhanov/australia
/home/ramielsarhanov/play
/home/ramielsarhanov/play/games
/home/ramielsarhanov/play/games/play
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.